

Dynamic Guideline: WHO-Based Dracunculiasis - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Scope note. WHO guidance for dracunculiasis / Guinea worm disease is primarily **eradication, surveillance, containment, water-safety, and community-mobilization guidance**, not drug-treatment guidance. WHO states that dracunculiasis is caused by *Dracunculus medinensis*, transmitted through unsafe drinking water containing infected copepods, has a 12–14 month incubation period, and has **no curative medicine or vaccine** [WHO EB128/15](#).

Exactly 5 WHO guidance documents synthesized for this report

#	WHO guidance document	Publication / adoption date used in this report	Target population	Scope and objectives
1	WHA78.14: Accelerating the eradication of dracunculiasis	27 May 2025	Member States with endemic, at-risk, or certification-relevant populations; eradication programmes	Calls for accelerated eradication, cross-border coordination, safe water, hygiene education, partner coordination, and action on animal infections, emergencies, and cross-border movement WHO WHA78.14 .
2	EB156(23): Accelerating the eradication of dracunculiasis	11 February 2025	Endemic Member States, national Guinea Worm Eradication Programmes, surveillance systems	Endorses strategy including community surveillance, animal tethering, aquatic-waste burial, filters, temephos, safe water, laboratory confirmation, rewards, immediate reporting, and monthly reporting from endemic areas WHO EB156(23) .

#	WHO guidance document	Publication / adoption date used in this report	Target population	Scope and objectives
3	EB156/CONF./8: Accelerating the eradication of dracunculiasis	11 February 2025	WHO governing bodies, endemic-country programmes, certification bodies	Summarizes eradication progress, including decline from 3.5 million estimated human cases in 1986 to 13 provisional cases in 2024, and notes the 2023 revised eradication definition requiring absence of adult female worm emergence in humans and animals for ≥3 consecutive years globally WHO EB156/CONF./8 .
4	Guidelines: Health Education and Community Mobilization in Dracunculiasis Eradication Programs	13 January 2014	Endemic communities, health educators, village volunteers, national eradication programmes	Provides community-level behavioural guidance: prevent persons with emerging worms/blisters/ulcers from entering drinking-water sources, promote household and community use of cloth filters, and mobilize communities for eradication activities WHO-hosted guideline PDF .
5	WHO Guidelines for drinking-water quality, fourth edition / rolling revision framework	2011 edition; 2022 rolling-revision PDF identified	Water-safety authorities, public-health programmes, communities dependent on unsafe water	Recognizes guinea worm / <i>Dracunculus medinensis</i> as a drinking-water-related microbial hazard and supports the principle that infective helminth stages should be absent from drinking-water WHO IRIS PDF , pS-Eau WHO PDF mirror .

Most recent WHO guideline-development-methodology item identified

The most recent WHO item found with the requested title pattern is “**Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)**”, launched through a WHO webinar scheduled for **11 December 2025**. It is **not a dracunculiasis guideline**, but it is relevant to WHO guideline-development methodology because it describes how youth engagement can strengthen guideline processes through lived experience, equity, inclusivity, and participation [WHO publication page](#), [WHO webinar launch page](#).

1.2 Key Recommendations

WHO-based clinical and public-health recommendations with GRADE ratings

Recommendation	GRADE Evidence Quality	Strength
Diagnose suspected dracunculiasis clinically when a compatible painful blister, ulcer, or emerging worm is identified in a person from an endemic or at-risk area; preserve and submit worm specimens for laboratory confirmation. WHO's current strategy includes laboratory confirmation of Guinea worm specimens WHO EB156(23) .	Moderate — clinical diagnosis is straightforward when the worm emerges; early/prepatent diagnosis remains unavailable.	Strong
Immediately report every suspected human case and animal infection; endemic areas should report at least monthly. WHO calls for human and animal dracunculiasis to be immediately reportable with at least monthly reporting from endemic areas WHO EB156(23) .	Moderate — based on consistent surveillance programme evidence and eradication-endgame necessity.	Strong
Contain every human case by preventing the patient from entering any water source after worm emergence; provide local wound care and supervised worm removal rather than drug cure. WHO states no curative medicine or vaccine exists WHO EB128/15 , and community guidance advises persons with emerging worms/blisters/ulcers not to enter drinking-water sources WHO-hosted guideline PDF .	Moderate — very strong biological plausibility and programme experience; no RCTs.	Strong
Do not use antiparasitic drug therapy as curative treatment outside research or expert-directed protocols; provide supportive wound care, pain control, infection prevention, and gradual worm extraction. Current WHO and review evidence state there is no available curative medicine or vaccine WHO EB128/15 , PMC review .	High for absence of approved curative therapy; Low for comparative effectiveness of supportive-care components.	Strong
Provide safe drinking-water sources to affected communities and treat drinking water through cloth/pipe filters and education. WHO endorses distribution and use of cloth and pipe filters and safe water provision WHO EB156(23) .	Moderate — long-standing eradication programme evidence and direct interruption of known transmission route.	Strong
Apply temephos monthly to unsafe drinking-water sources in endemic communities where indicated. WHO's endorsed strategy includes monthly temephos application to unsafe drinking-water sources in endemic communities WHO EB156(23) .	Moderate — biologically direct vector-control mechanism and extensive programme experience.	Strong
Implement health education and community mobilization to prevent contamination of drinking-water sources and promote correct filter use. WHO-hosted community guidelines emphasize repeated demonstrations and household/community behaviour change WHO-hosted guideline PDF .	Moderate — supported by eradication programme experience; limited controlled-trial evidence.	Strong

Recommendation	GRADE Evidence Quality	Strength
In endemic and at-risk communities, proactively tether domestic animals, prevent infected animals from entering water, reward reporting/tethering, and bury aquatic waste such as fish entrails. WHO endorses animal tethering, aquatic-waste burial, and rewards for infected-animal reporting/tethering WHO EB156(23) .	Moderate — animal infections are now central to persistence; evidence mainly observational/programmatic.	Strong
Intensify cross-border surveillance, joint investigations, and information-sharing where populations, animals, or water sources cross administrative borders. WHA78.14 calls for intensified cross-border collaboration, joint surveillance, and information sharing WHO WHA78.14 .	Low to Moderate — operational evidence and plausibility; difficult to test experimentally.	Strong
For certification, document absence of adult female worm emergence in both humans and animals for at least three consecutive years at the relevant certification level. WHO notes the 2023 revised global eradication definition includes humans and animals WHO EB156/CONF./8 .	Moderate — certification standard based on epidemiology and expert consensus.	Strong

1.3 Guideline Development Methodology

Evidence Review Process

- The WHO dracunculiasis guidance base is derived mainly from:
 - WHO governing-body resolutions and decisions on eradication strategy [WHO WHA78.14](#), [WHO EB156\(23\)](#).
 - WHO Secretariat reports summarizing disease biology, transmission, lack of drug/vaccine, and eradication interventions [WHO EB128/15](#).
 - WHO-hosted operational guidance for health education and community mobilization [WHO-hosted guideline PDF](#).
 - WHO drinking-water-quality guidance recognizing *Dracunculus medinensis* as a drinking-water-related hazard [WHO IRIS PDF](#).
 - Surveillance and certification evidence from WHO, CDC/MMWR, and eradication partners [CDC MMWR](#).

Consensus Method

- WHO recommendations are largely produced through **Member State governance processes**, WHO Secretariat synthesis, expert eradication bodies, and operational programme consensus.
- The International Commission for the Certification of Dracunculiasis Eradication revised the definition of worldwide eradication in 2023 to require confirmed absence of adult female worm emergence in **humans and animals** for ≥3 consecutive years globally [WHO EB156/CONF./8](#).
- WHO’s broader guideline-development methodology increasingly emphasizes inclusivity and meaningful engagement; the WHO youth-engagement case study for the abortion care guideline

illustrates how lived experience and youth participation can enrich guideline development, although this was not specific to dracunculiasis [WHO publication page](#).

Conflict of Interest Management

- The dracunculiasis WHO governing-body documents in the supplied evidence do not provide a detailed disease-specific conflict-of-interest table.
- For this report, recommendations were graded transparently using a GRADE-style framework:
 - **High**: very confident evidence or universally accepted factual status.
 - **Moderate**: consistent observational/programmatic evidence with strong biological plausibility.
 - **Low**: limited observational evidence, operational consensus, or indirect evidence.
 - **Very Low**: preclinical, animal-only, uncontrolled, or highly indirect evidence.
- Because dracunculiasis eradication interventions are community and public-health interventions, most evidence is **observational, ecological, and programme-based**, not randomized.

II. Evidence Summary After WHO Latest Guideline Release (Dracunculus: background document for the WHO guidelines for drinking-water quality) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy

Evidence reviewed for the post-guideline update included WHO, CDC/MMWR, WHO AFRO, peer-reviewed reviews, trial-related literature, and clinical-trial registry signals through **29 April 2026**. Priority was given to sources addressing:

- Diagnosis and laboratory confirmation.
- Human case counts and containment.
- Animal infections and zoonotic transmission.
- Water-safety, filtration, temephos, and community mobilization.
- Drug/vaccine/diagnostic-assay development.
- Certification and eradication definitions.

Databases and Sources Searched

- WHO official documents and WHO-hosted PDFs.
- WHO AFRO technical and progress reports.
- CDC/MMWR surveillance updates.
- Peer-reviewed review literature in PubMed Central and journal sources.
- ClinicalTrials.gov search-result signals.
- Partner technical sources where they corroborated WHO/CDC data.

Inclusion Criteria

Included evidence if it:

1. Addressed dracunculiasis / Guinea worm disease / *Dracunculus medinensis*.

- 2. Reported diagnosis, treatment, case management, surveillance, eradication, water safety, animal infections, or certification.
- 3. Came from WHO, CDC/MMWR, peer-reviewed literature, or an authoritative eradication partner.
- 4. Was current or relevant to the eradication endgame.

Exclusion Criteria

Excluded evidence if it:

- 1. Was not disease-specific.
- 2. Did not contain extractable findings.
- 3. Was unrelated to diagnosis, treatment, management, surveillance, or eradication.
- 4. Was a non-authoritative secondary summary without corroboration.

2.2 Key New Studies

Study / source	Design	Key Findings	GRADE Quality
CDC MMWR: “Progress Toward Eradication of Dracunculiasis — Worldwide, January 2024–June 2025”	Global surveillance report	In 2024, 15 human cases were reported worldwide: 9 in Chad and 6 in South Sudan. In January–June 2025, 1 human case was reported versus 3 in January–June 2024. Animal infections remained much more frequent: 664 animal infections in 2024 and 550 in January–June 2025 CDC MMWR , CDC PDF .	Moderate
CDC MMWR animal-infection analysis, January 2024–June 2025	Surveillance subgroup analysis	Animal infections were concentrated in Cameroon and Chad. In 2024, Cameroon and Chad accounted for 591/664 animal infections ; in January–June 2025, they accounted for 478/550 animal infections CDC PDF .	Moderate
WHO EB156/CONF./8 and WHA78.14 eradication update	WHO governance and programme synthesis	Human cases declined from an estimated 3.5 million in 1986 to 13 provisional cases in 2024 ; WHO emphasizes animal infections, unsafe water, health/humanitarian emergencies, and cross-border movement as persistence factors WHO EB156/CONF./8 , WHO WHA78.14 .	Moderate

Study / source	Design	Key Findings	GRADE Quality
WHO IRIS: Fortieth Meeting of the International Task Force for Disease Eradication, 28–29 October 2025	Expert meeting summary	Reviewed the status of dracunculiasis eradication and included a signal of 10 human cases reported in 2025 in the supplied summary WHO IRIS PDF .	Low to Moderate
PMC review on neglected tropical diseases	Narrative/review evidence	Diagnosis remains primarily clinical based on recognition of an emerging worm; microscopy for larvae may be used. There is no early-stage diagnostic tool ; antigen-detection assays for prepatent infection are a future need PMC review .	Low
MDPI Tropical Medicine and Infectious Disease review, 2022	Narrative review	Reported 27 human cases in 2020 and 14 human cases in 2021 , and highlighted persistent eradication challenges including reaching rural areas and controlling infection sources MDPI Tropical Medicine and Infectious Disease .	Low
WHO AFRO progress report 2020–2022	Regional programme report	Reported 13 human cases in 2022 , the lowest annual figure since eradication efforts began in 1986, and 188 Member States certified free of Guinea worm transmission WHO AFRO progress report .	Moderate
Flubendazole study in laboratory-reared ferrets and domestic dogs in Chad	Animal clinical trials / veterinary research	Subcutaneous flubendazole produced contrasting results; larvae from treated gravid female worms in ferrets had poor condition, low-to-no motility, and inability to infect copepods, but evidence does not establish human treatment efficacy PMC flubendazole study .	Very Low for human recommendations; Low for animal-research signal
ClinicalTrials.gov search signal	Trial-registry search	No verifiable ClinicalTrials.gov-registered human dracunculiasis vaccine, drug, or diagnostic-assay trial was identified in the supplied evidence ClinicalTrials.gov .	Low

2.3 Evidence Gaps and Future Research Needs

1. Early diagnosis

- Current diagnosis is usually late, when the worm emerges.
- There is no validated early-stage diagnostic tool; antigen-detection assays for prepatent infection are a stated research need [PMC review](#).

2. **Animal reservoirs**

- Animal infections, especially in dogs, are now the major eradication barrier.
- 2024–2025 CDC data show animal infections far exceed human cases [CDC MMWR](#).

3. **Transmission pathways involving aquatic animals**

- Evidence increasingly supports the need to prevent consumption of raw/undercooked aquatic animals or fish entrails by dogs and other animals.
- WHO-endorsed aquatic-waste burial should be strengthened and evaluated [WHO EB156\(23\)](#).

4. **Drug and vaccine development**

- There is no available curative medicine or vaccine [WHO EB128/15](#).
- Flubendazole and other veterinary or antiparasitic approaches remain investigational and cannot yet change human management recommendations [PMC flubendazole study](#).

5. **Security and access**

- Civil unrest and insecurity in countries such as Mali, South Sudan, and Sudan threaten surveillance quality and eradication timelines [CDC PDF](#).

6. **Certification in humans and animals**

- The 2023 revised eradication definition means programmes must demonstrate absence of adult female worm emergence in both humans and animals for ≥3 consecutive years globally [WHO EB156/CONF./8](#).

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
Maintain community-based surveillance in endemic and at-risk communities.	Human cases remain extremely low, but ongoing cases in 2024–2025 confirm that surveillance must continue until certification criteria are met CDC MMWR .	Continue active community-based surveillance, rumor investigation, immediate reporting, and monthly endemic-area reporting until certification is achieved.
Laboratory-confirm Guinea worm specimens.	Diagnosis remains largely clinical, and early diagnostics are lacking; laboratory confirmation remains essential for certification PMC review , WHO EB156(23) .	Submit all human and animal worm specimens for laboratory confirmation; prioritize development and field validation of early diagnostic assays.

Original Recommendation	New Evidence Impact	Updated Recommendation
Prevent persons with emerging worms from entering water.	No new evidence supports drug cure; water contamination by emerging worms remains central to transmission.	Immediately isolate and support patients with emerging worms so they do not enter water sources; provide wound care, pain control, and supervised gradual extraction.
Provide filters, safe water, and health education.	No evidence weakens this recommendation; ongoing transmission in remote communities supports intensification.	Maintain universal access to safe water in affected communities; distribute and demonstrate correct use of cloth/pipe filters; reinforce health education repeatedly.
Apply temephos to unsafe water sources.	Continued low-level transmission supports targeted vector control in endemic communities.	Apply temephos monthly to unsafe drinking-water sources in endemic communities when operationally appropriate and safe.
Address animal infections through tethering and reporting.	Animal infections remain much more frequent than human cases; Cameroon and Chad account for most recent animal infections CDC PDF .	Escalate animal-infection interventions: proactive dog tethering, rewards for reporting/tethering, prevention of animal access to water, and burial of aquatic waste/fish entrails.
Intensify cross-border collaboration.	Ongoing transmission and population/animal movement continue to threaten eradication WHO WHA78.14 .	Implement joint cross-border surveillance, shared line lists, synchronized investigations, and coordinated water-source interventions.
Certify eradication after absence of disease.	Certification now explicitly includes humans and animals for ≥ 3 consecutive years globally WHO EB156/CONF/8 .	Use human-and-animal surveillance data for certification; do not consider human-case interruption alone sufficient for eradication certification.

Practical WHO-Based Management Algorithm

1. Suspect

- Painful blister, ulcer, or visible emerging worm in a person from an endemic or at-risk area.
- Suspected infected dog, cat, baboon, or other animal with emerging worm in an endemic area.

2. Immediate actions

- Prevent the person or animal from entering any water source.
- Notify the local Guinea Worm Eradication Programme immediately.
- Begin case containment and community investigation.

3. Clinical care

- Provide local wound care.
- Gradually extract the worm using standard supervised technique.
- Treat secondary bacterial infection if clinically present.
- Provide pain control and functional support.
- Educate patient/family not to contaminate water.

4. Public-health response

- Investigate source village and water exposures.
- Provide or reinforce cloth/pipe filter use.
- Treat unsafe water sources with temephos where indicated.
- Ensure safe water access.
- Search for additional human and animal cases.
- Submit worm specimens for laboratory confirmation.

5. Animal-response package

- Tether infected animals.
- Prevent animal access to water.
- Reward reporting and tethering where programme policy allows.
- Bury aquatic waste/fish entrails to prevent animal consumption.

6. Certification-readiness

- Maintain high-quality surveillance for humans and animals.
- Document absence of adult female worm emergence.
- Sustain surveillance for at least three consecutive years before certification review.

Final Evidence-Based Conclusion

As of **29 April 2026**, WHO-based dracunculiasis management remains an **eradication-endgame strategy**, not a drug-treatment strategy. The most important actionable measures are: rapid clinical recognition, laboratory confirmation, immediate reporting, case containment, safe water, filtration, temephos vector control, community education, animal tethering, aquatic-waste burial, rewards for reporting, and cross-border surveillance. Human cases are now extremely rare, but animal infections—especially in dogs and concentrated in Cameroon and Chad—remain the principal threat to eradication [CDC MMWR](#), [WHO EB156\(23\)](#).